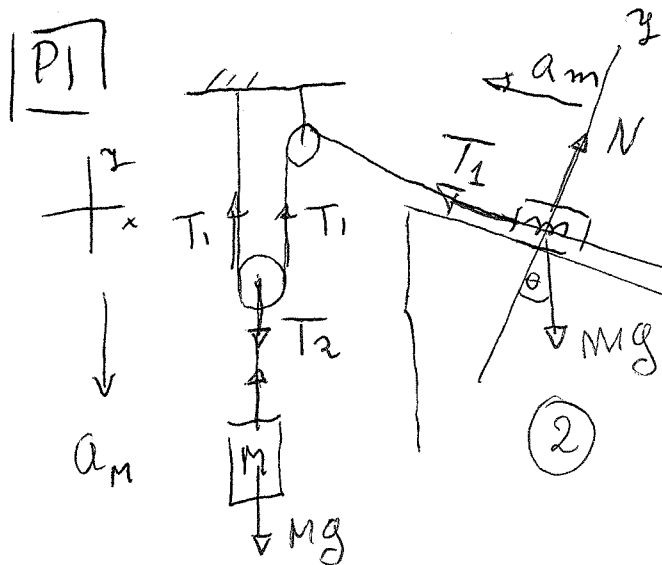




$$\underline{M} \quad (1) \sum F_y: Mg - T_2 = M a_m \quad (1)$$

$$\underline{P} \quad (2) \sum F_y: T_2 - 2T_1 = 0 \quad 0,5$$

$$\underline{m} \quad (3) \sum F_x: T_1 - mg \sin \theta = m a_m \quad (1)$$

$$(4) \sum F_y: N - mg \cos \theta = 0 \quad 0,5$$

$$(5) \quad a_m = 2a_m$$

$$(1) + (2) \text{ usando } (5) \quad Mg - 2T_1 = M a_m / 2$$

$$\text{de } (3) \quad T_1 = m a_m + mg \sin \theta$$

$$\text{queda} \quad Mg - 2(m a_m + mg \sin \theta) = M \frac{a_m}{2}$$

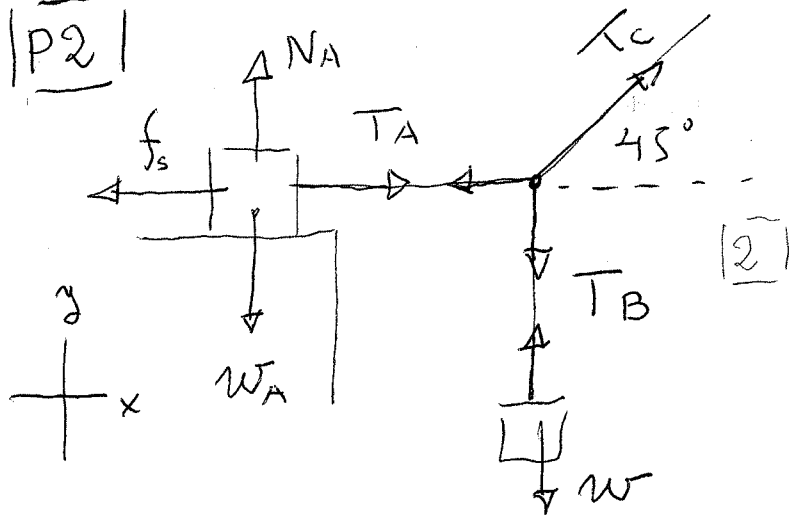
$$2mg - 2ma_m - 2mg \sin \theta = \cancel{2m} a_m / \cancel{2}$$

$$2g(1 - \sin \theta) = a_m(1 + 2)$$

$$a_m = \frac{2}{3} g (1 - \sin \theta) //$$

$$a_m = \frac{g}{3} (1 - \sin \theta) //$$

P2



$$\begin{aligned} \text{M}_A \quad (1) \quad \sum F_x: T_A - f_s &= 0 \\ (2) \quad \sum F_y: N_A - W_A &= 0 \end{aligned}$$

$$\text{W} \quad (3) \quad \sum F_y: W - T_B = 0$$

$$\text{NUDO} \quad (4) \quad \sum F_x: T_C \cos 45^\circ - T_A = 0$$

$$(5) \quad \sum F_y: T_C \sin 45^\circ = T_B$$

70%
4,9

(a) de (1) $f_s = T_A$ *

(1) de (4) $T_A = T_C \cos 45^\circ$

de (3) y (5) $T_C = W / \sin 45^\circ$

(4) queda $T_A = \left(\frac{W}{\sin 45^\circ} \right) \cos 45^\circ$ en *

$$f_s = W / \tan 45^\circ = 15 \text{ (N)} //$$

(b) Para el valor máximo $f_s = \mu_s N_A$

$$\mu_s N_A = \frac{W_{\max}}{\tan 45^\circ}$$

$$\tan 45^\circ = 1$$

(1) de (2) $N_A = W_A = 90,0 \text{ [N]}$

$$\mu_s W_A = \frac{W_{\max}}{\tan 45^\circ}$$

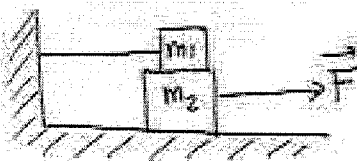
$$W_{\max} = \mu_s W_A \tan 45^\circ$$

$$W_{\max} = 0,3 \times 90 \times 1 \text{ [N]}$$

$$W_{\max} = 27 \text{ [N]} //$$

Prueba 3

Ejercicio 3

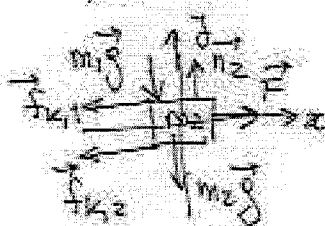
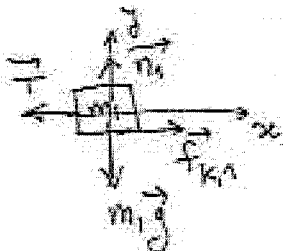


$$F = 46,0 \text{ N}$$

$$m_1 = 5,00 \text{ N}$$

$$m_2 = 10,0 \text{ N}$$

$$\mu_k = 0,200$$



$$\begin{aligned} (1) \quad & -T + f_{k1} = 0 \\ (2) \quad & -n_1 + m_1 g = 0 \end{aligned}$$

$$\begin{aligned} (3) \quad & F - f_{k1} - f_{k2} = m_2 a \\ (4) \quad & n_2 - m_1 g - m_2 g = 0 \end{aligned}$$

$$(5) \quad f_k = \mu_k n$$

$$\text{de (2)} \quad n_1 = m_1 g \Rightarrow f_{k1} = \mu_k m_1 g \quad (6)$$

$$\text{de (4)} \quad n_2 = (m_1 + m_2) g \Rightarrow f_{k2} = \mu_k (m_1 + m_2) g \quad (7)$$

$$\text{de (1)} \quad T = f_{k1} \xrightarrow{\text{con (6)}} \boxed{T = \mu_k m_1 g}$$

$$T = 0,200 \cdot 5,00 \text{ kp} \cdot 9,8 \text{ m/s}^2 \rightarrow \boxed{T = 9,8 \text{ N}}$$

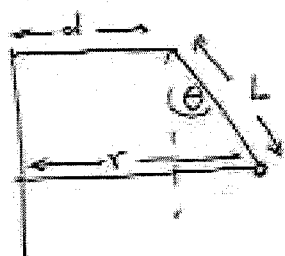
$$\text{de (3)} \quad a = \frac{F - f_{k1} - f_{k2}}{m_2} = \frac{F - \mu_k m_1 g - \mu_k (m_1 + m_2) g}{m_2} \quad (6), (7)$$

$$a = \frac{F - \mu_k (2m_1 + m_2) g}{m_2} \Rightarrow a = \frac{46,0 \text{ N} - 0,200 (2 \cdot 5,00 \text{ kp} + 10,0 \text{ kp}) 9,8 \text{ m/s}^2}{10,0 \text{ kp}}$$

$$\boxed{a = 0,08 \text{ m/s}^2}$$

Prueba 3

Ejercicio 4



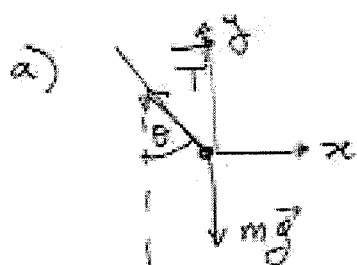
$$\theta = 30,0^\circ$$

$$d = 3,00 \text{ m}$$

$$L = 5,00 \text{ m}$$

$$r = d + L \cos \theta = 3,00 \text{ m} + 5,00 \text{ m} \cdot \cos 30^\circ$$

$$r = 5,50 \text{ m}$$



$$\begin{cases} T \sin \theta = m a_c & (1) \\ T \cos \theta - m g = 0 & (2) \end{cases}$$

$$a_c = \frac{v^2}{r} = \omega^2 r \quad (3)$$

Dividiendo miembro a miembro: eq. (1) y (2):

$$T \sin \theta = m a_c$$

$$T \cos \theta = m g$$

$$\tan \theta = a_c / g$$

$\Rightarrow$

$$\tan \theta = \frac{a_c}{g} \overset{\text{usando (3)}}{=} \frac{1}{g} \frac{v^2}{r} \Rightarrow$$

$$\Rightarrow v^2 = g \cdot r \cdot \tan \theta \Rightarrow v = \sqrt{g \cdot r \cdot \tan \theta}$$

$$v = \sqrt{9,8 \text{ m/s}^2 \cdot 5,50 \text{ m} \cdot \tan 30^\circ} = 5,58 \text{ m/s}$$

$$P = \frac{2\pi r}{v} \Rightarrow$$

~~2\pi r~~

$$P = \frac{2\pi \cdot 5,50 \text{ m}}{5,58 \text{ m/s}} \Rightarrow$$

$$P = 6,13 \text{ s}$$

b) en la parte a) se obtuvo

$$\tan \theta = \frac{1}{g} \frac{v^2}{r}$$

no depende de la masa del pasajero.